

Scenario: Library Management System

The Library Management System is a basic system that allows users (students and librarians) to manage book borrowing and returns. In this system, the main activities include Registration, login searching for books, borrowing books, and returning books. The system also lets librarians manage book inventory. And both can Logout.

1. Use Case Diagram

The primary actors in this scenario are:

- **Student:** Can self-register, log in, search for books, borrow books, return books and logout.
- **Librarian:** Can login, manage book inventory, assist with book borrowing and returning and logout.

Use Cases:

- **Register:** A user registers themselves in the system.
- **Login:** Both students and librarians can log in to the system.
- **Search Book:** Any registered user can search for a book.
- **Borrow Book:** Users can borrow books if available.
- **Return Book:** Users can return borrowed books.
- **Manage Book Inventory:** Librarians can add, update, and remove books from the system.
- **Logout:** Both students and librarians can log out of the system.

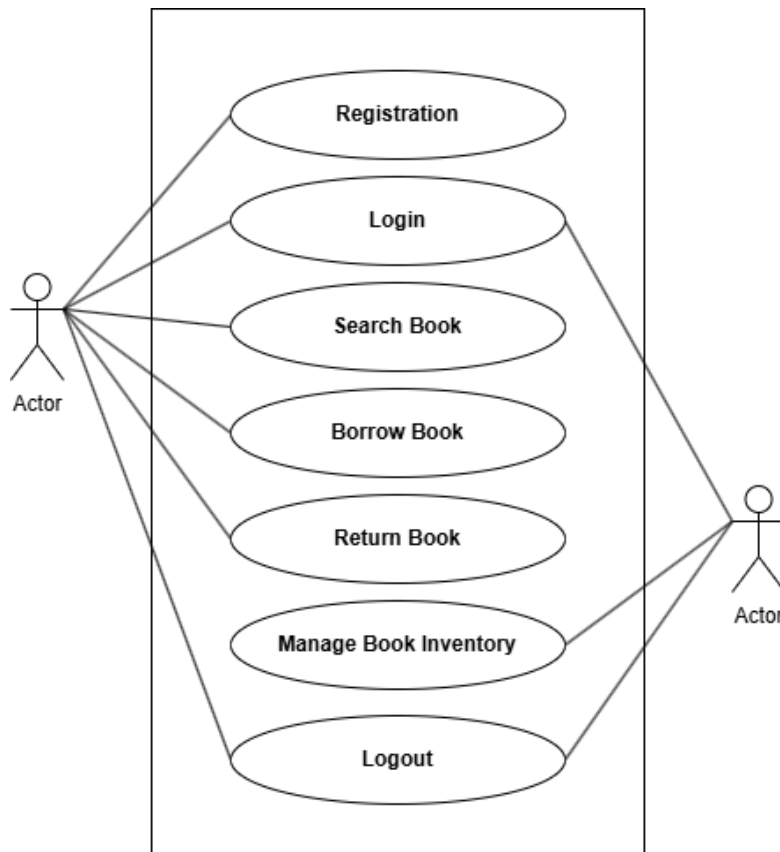


Figure-: Use Case Diagram for Library Management System

User Case Description

Case Description-01: Registration

Use Case	Registration	
Goal	Student must be logged in.	
Precondition	Users must have access for registration.	
Success End Condition	Notification: “Book Borrowed Successfully.”	
Failed End Condition	Notification: “Book is Unavailable.”	
Primary Actors:	Student	
Secondary Actors:		
Trigger	Student initiates a borrowing request for a specific book.	
Description / Main Success Scenario	1.	Student initiates a borrowing request for a specific book.
	2.	System checks book availability
	3.	Enter Information
	4.	If available, system registers the book as borrowed
	5.	System confirms the borrowing
	6.	The system saves the details and shows them !!!Successfully Booked!!! Notify
Alternative Flows	1.1	System Error
		1.1.a. Try Again!!
	2.1	System Doesn't work.
		2.1.a. Try Again Later!
	4.1	The user Did not fill up the details!
		4.1.a. Checked By the system & Notify by “Please! Fill Up the Box”.
	5.1	The system did not respond
		5.1.a. Show Error Message.
	6.1	The system Doesn’t save the details.
		6.1.a. Notification: “Details did not Save”
Quality Requirements	The user Will fill up all the details in 30 minutes.	

Case Description-02: Login

Use Case	Login	
Goal	Users can enter the system by the login.	
Precondition	Users must have to be registered First.	
Success End Condition	Notification: “Login Successful”	
Failed End Condition	Notification: “Login Failed!!”	
Primary Actors:	Student, Librarian	
Secondary Actors:		
Trigger	The user will request a login to enter the system	
Description / Main Success Scenario	1.	Press “Login” Button
	2.	Provide login interface
	3.	Enter user id and password.
	4.	User provide the “Login” button
	5.	Verified and login
	6.	Notification: “Login Successful”
Alternative Flows	1.1	System Error
		1.1.a. Try again
	2.1	Server Not Found
		2.1.a. Try Again Later!
	4.1	The system Did not respond.
		4.1.a. Show error message.
	5.1	Information Error!!
		5.1.a. Notification: “Enter the right User ID and Password.”
Quality Requirements	Users fill up the login info within 10 minutes.	

Case Description-03: Search Book

Use Case	Search Book	
Goal	The user searches for books by title, author, or genre in the system.	
Precondition	Must visit Search Box.	
Success End Condition	Notification “Here is the list:”	
Failed End Condition	Notification “Sorry, Nothing Found!”.	
Primary Actors:	Student	
Secondary Actors:		
Trigger	Users will search for books by title, author, or genre in the system.	
Description / Main Success Scenario	1.	Visit Dashboard
	2.	Press “Search”
	3.	Provide Search Box
	4.	Fill Search Box
	5.	User press “Search” button
	6.	Search Result shown as list.
Alternative Flows	2.1	System Error
		2.1.a Try again
	3.1	Not responding
		3.1.a Try later
	5.1	Not Found
		5.1.a Notification: “Sorry, Nothing Found!”
Quality Requirements	Search will be done by keywords	